THE CHINESE ECONOMY

A JOURNEY INTO HISTORY, ECONOMICS, AND POLICY

Lecture 7: Reforming the state firms

TODAY

- **Your wedges**: the homework, aloud
- **Hsieh and Klenow in one picture**: the SOE Lab again
- **What China did with its state firms**: Hsieh and Song (2015), in seven steps
- **A wedge that disappeared overnight**: Khandelwal, Schott and Wei (2013), in seven steps
- **Three ways to see misallocation**

YOUR WEDGES

THE HOMEWORK, ALOUD

On capital

some firms borrow cheaper, or dearer, than others

On size

rules, taxes or subsidies that make some firms stay small, or grow too big

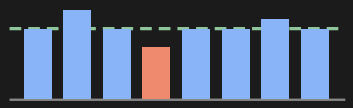
On entry

a licence, a permit, a connection: some firms may not enter at all

- For each wedge: **who gains, who loses**, what it costs the economy
- Is it being **removed**, or getting **stronger**?

THE SAME THREE QUESTIONS, ALL DAY

In Hsieh-Klenow's language

Which wedge?	*On capital, on size, or on entry? Who gains, who loses?*
Which marginal product does it bend?	*MRPK, MRPL, or both? Up or down?*
What happens to the dispersion of TFPR?	*Which bars leave the line, and which way?* 

Bars: TFPR of each firm (blue private, red state). Dashed line: where all of them would be without wedges

HSIEH AND KLENOW IN ONE PICTURE

- All firms have the **same technology**; they differ in how good they are (TFPQ)
- Without wedges, capital and workers move until each one earns the **same** everywhere: same MRPK, same MRPL, so the **same TFPR**
- A wedge keeps a firm **too big or too small**: its TFPR moves off the line
- The loss: the output we could have with the **same** capital and workers
- **SOE Lab**: preset 1998, look at the bars, press "Reallocate". Then open "Production" under the lab

TWO TFPS, IN TWO FIRMS

LECTURE 6, ONCE MORE, WITH NUMBERS

	TFPQ	Capital and workers	Units sold	Price	Revenue	TFPR = price × TFPQ
Firm A	1	1	1	1	1	**1**
Firm B, no wedge	2	4	8	½	4	**1**
Firm B, kept small	2	2	4	0.63	2.5	**1.26**

- Every firm sets its price the **same percentage** above its cost. B's cost is half, so B's price falls as it grows
- With no wedge, B grows until its revenue per unit of input is the same as A's: TFPQ **different**, TFPR **the same**
- Keep B small, and it sells little at a high price: its TFPR is **higher**. A gap in TFPR is the sign of a wedge
- Khandelwal, Schott and Wei read it backwards: same percentage above cost, so a **lower price** means a **better** firm

GRASP THE LARGE, LET GO OF THE SMALL

Chang-Tai Hsieh, Zheng (Michael) Song

Brookings Papers on Economic Activity, 2015

- The title is the slogan of a reform the Communist Party announced in **1999**: close or sell the **small** state firms, merge the **big** ones into groups the state controls
- The machinery of Hsieh and Klenow, on the same data, up to 2007 and 2012

MOTIVATION

- 1998: state firms have about **half** of industrial revenue, 60% of the capital, and **4 in 10** lose money
- Story one, **"markets over Mao"**: China grew because the state let its firms die or go private
- Story two, **"state capitalism"**: 67 of the 69 Chinese firms in the 2014 Fortune Global 500 belong to the state. "The state advances, the private sector retreats"
- Both stories are told **without evidence**

QUESTION

HSIEH AND SONG (2015) · 2. QUESTION

What did China do with its state firms after 1998, and how much of its output does that explain?

DATA

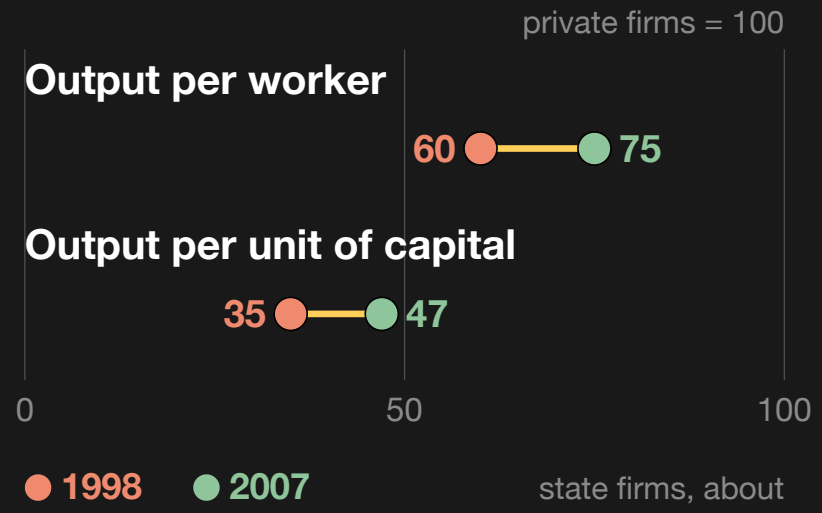
- The census of Chinese industrial firms, **1998-2007 and 2012**: the data of Hsieh and Klenow, extended
- The hard part: **finding the state**. A firm is state-owned if the state holds at least half of its capital or controls it
- They checked by hand the biggest 1% of firms in 2007: two thirds are controlled by the state, and almost half of these are **registered as private**

EMPIRICAL STRATEGY

- The model of Hsieh and Klenow: for each firm, its **TFPQ** and its **wedges** on capital and labour
- Follow the state firms of 1998: **closed**, **privatized**, or **kept**? And the new ones?
- Then the "what if": start from 2007, give the reformed firms back their **1998** TFPQ and wedges, remove the new state firms. How much lower is output?
- Two rules of the world decide the answer: is capital available at a **world** interest rate? Did state firms have **surplus** workers?

RESULTS: THE FACTS

HSIEH AND SONG (2015) · 5. RESULTS



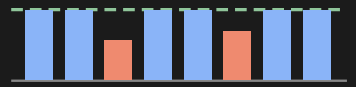
State firms against private firms. Hsieh and Song, Tables 4-5

- By 2007, **more than 8 in 10** of the state firms of 1998 were closed or privatized
- The survivors cut **3.6 million** workers: output per worker catches up
- Output per unit of capital catches up **much less**, and privatized firms look like state firms
- About a third of the state firms of 2007 are **new**: big, and heavy on capital

RESULTS, IN HSIEH-KLENOW'S LANGUAGE

In Hsieh-Klenow's language

Which wedge?	Cheap credit for state firms, on capital. Too many workers kept on, on labour
Which marginal product does it bend?	Output per worker of state firms: from 60% to 75% of private firms, as they shed surplus workers (it is the MRPL only if no worker is surplus). MRPK: from 35% to only 47%, because capital still comes on easy terms
What happens to the dispersion of TFPR?	The labour gap closes a lot, the capital gap much less. Part of the gain is still on the table



Bars: TFPR of each firm (blue private, red state). Dashed line: where all of them would be without wedges

RESULTS: THE SURPRISE

HSIEH AND SONG (2015) · 5. RESULTS

Can making a state firm more productive make the economy produce less?

- Yes. If the firm earns **less** than average on its capital and workers, a better firm pulls **more** of them in, to where they produce least
- You saw it yesterday, **SOE Lab 3**: "Upgrade state firms +30%" gives +15% directly, and the inputs moving to them take away about 8%
- So a reform of bad firms is good only if the **wedges** go too

RESULTS: WHAT THE REFORM WAS WORTH

HSIEH AND SONG (2015) · 5. RESULTS

How much higher 2007 output is, because of the reform (%)

	Open economy	Open, surplus workers	Closed economy	Closed, surplus workers
"Let go of the small"	1.6	3.2	0.0	0.9
"Grasp the large"	10.0	18.4	-3.5	0.9
Both	11.4	**21.0**	-3.0	1.9

- 21% is a **level**: 2007 output, compared with no reform. The abstract calls it "21 percent of China's growth"
- With a **closed** economy the gain almost disappears, or turns negative: better state firms pull scarce capital to where it earns least

ROBUSTNESS

HSIEH AND SONG (2015) · 6. ROBUSTNESS, OR WHAT A REFEREE WOULD ASK

- **Wedge or technology?** The model gives state and private firms the same technology. Brandt: "highly restrictive"; with a more flexible one, state firms' productivity grows **less**
- **Open or closed?** The sign of the result depends on it
- **Where is the counterfactual?** No experiment: the government chose whom to close, and the "what if" comes from the model
- **The data**: no small firms, four missing years, and no value added in 2012

WHY IT MATTERS

- Neither story is right. Closing and selling state firms added **little**; reforming the ones the state kept added **about a fifth**; private firms did the **rest**
- Reforming bad firms is not enough: what counts is **where** capital and workers go
- But the model cannot tell us what **caused** what. For that we need a wedge that was removed by someone else, on a known day

TRADE LIBERALIZATION AND EMBEDDED INSTITUTIONAL REFORM

Amit K. Khandelwal, Peter K. Schott, Shang-Jin Wei

American Economic Review, 2013

- Read the title: removing a **trade barrier** is the cause; reforming the **institution** built around it comes with it

MOTIVATION

- Until **1 January 2005**, China's textile and clothing exports to the US, the EU and Canada had **quotas**
- The quotas were set **abroad**. Who could use them was decided in **Beijing**: the government gave out export licences
- Officially by past exports. In practice also by **politics**, and selling a licence was illegal
- So removing the quota may remove **two** costs: the barrier, and the institution that managed it

QUESTION

KHANDELWAL, SCHOTT AND WEI (2013) · 2. QUESTION

Did the government give the export licences to its best firms? If not, what did it cost?

DATA

- Chinese **customs** data, 2000-2005: for each firm, product and destination, the value and the quantity exported, so the **price**
- And who owns the firm: the **state**, Chinese private owners, or foreigners
- Goods **under quota** until 2004 (cotton slips to the US) and similar goods **never under quota** (silk slips to the US)

QUESTION

They have no TFP for most firms. How can they tell who is good?

- From the **price**. Remember lecture 6: every firm charges the same markup, so price is always the **same multiple** of marginal cost
- So a lower price means a lower cost: a **better** firm, if quality is the same
- They also correct prices for quality: a good that sells more **at the same price** counts as better

EMPIRICAL STRATEGY

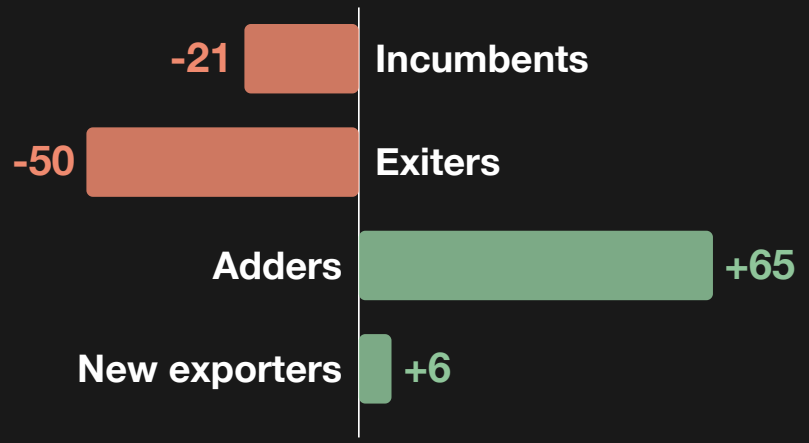
State firms' share of exports (%)

	2004	2005	Change
Under quota	54	38	-16
Never under quota	44	36	-8
Difference			-8

- A **difference in differences** (lecture 2): the goods never under quota show what would have happened anyway
- And a benchmark: if licences had gone to the **best** firms, the boom would come from the firms that **already** had them, the biggest first

RESULTS: WHO GREW

KHANDELWAL, SCHOTT AND WEI (2013) · 5. RESULTS



Change in market share, 2004 to 2005, points

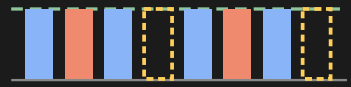
Goods under quota, average over product and destination. Adders: firms that already exported other goods. Khandelwal, Schott and Wei, Table 5

- In 2005, exports of quota goods jump **119%** (17% a year before) and the number of exporters almost **doubles**
- The growth comes from **entry**: firms new to that good and destination, mostly private
- Entrants' prices are **25% lower** than those of the firms that had the licences
- The firms with the **biggest** shares under quota lose the most

RESULTS, IN HSIEH-KLENOW'S LANGUAGE

In Hsieh-Klenow's language

Which wedge?	An export licence, handed out by the government, mostly to state firms. A firm without one cannot sell abroad: its wedge is infinite
Which marginal product does it bend?	The excluded firms would have high marginal products, but they cannot use them
What happens to the dispersion of TFPR?	Hsieh and Klenow cannot see it: the excluded firms (dashed) are not among the exporters. Remove the licences, and they enter with prices 25% lower



Bars: TFPR of each firm (blue private, red state). Dashed line: where all of them would be without wedges

RESULTS: WHAT IT WAS WORTH

KHANDELWAL, SCHOTT AND WEI (2013) · 5. RESULTS

- Removing the quotas raised exporters' productivity by about **7%**
- If the licences had gone to the best firms, the gain would have been smaller: the misallocated licences make it **28%** larger
- This number comes from a **model**, calibrated to the facts of the previous slides
- Textiles and clothing: **15%** of China's exports, **13%** of manufacturing jobs

ROBUSTNESS

KHANDELWAL, SCHOTT AND WEI (2013) · 6. ROBUSTNESS, OR WHAT A REFEREE WOULD ASK

- **Prices are not productivity**: same markup, quality corrected from demand. The same rule of the world as Hsieh and Klenow
- **One year**: in 2005 firms may have exported more to win a share of future quotas. The US and the EU did bring some back
- **Subcontracting**: if licence holders had good firms produce for them, the "entrants" were already producing
- **The control group**: most entrants are firms adding quota goods. If they moved production from the other goods, the difference is too big

WHY IT MATTERS

- Trade can pay more than the models say: opening up also kills the **institutions** that grew around the barrier
- A reform from **outside**: a date fixed abroad did what Beijing had not done
- Misallocation is also about **who is allowed in**. Entry, exit and the firms that never appear will matter even more when we turn to **growth**

THREE WAYS TO SEE MISALLOCATION

	What they do	What it costs them
Hsieh and Klenow	Measure the dispersion of TFPR, against the US	All wedges together; errors look like wedges
Hsieh and Song	Follow one reform through the same model	No experiment: the "what if" is the model's
Khandelwal, Schott and Wei	Watch one wedge disappear, on a known day	One sector, one year

The wider the view, the weaker the proof. The sharper the proof, the narrower the view